

# Dongfang Zhao, Ph.D.

Status: Male, U.S. Citizen, Married with two children

Position: Assistant Professor of Computer Science, University of Nevada, Reno

## Professional Preparation

|                                       |             |                       |               |
|---------------------------------------|-------------|-----------------------|---------------|
| Northeastern University               | China       | Computer Science      | BE, 2003      |
| Katholieke Universiteit Leuven        | Belgium     | Statistics            | MS, 2005      |
| Emory University                      | Atlanta, GA | Computer Science      | MS, 2008      |
| Illinois Institute of Technology      | Chicago, IL | Computer Science      | PhD, 2015     |
| Pacific Northwest National Laboratory | Seattle, WA | Applied Cryptography  | PostDoc, 2016 |
| University of Washington              | Seattle, WA | Distributed Databases | PostDoc, 2017 |

## Employment

|           |                                                                                   |
|-----------|-----------------------------------------------------------------------------------|
| 2017–Now  | Assistant Professor of Computer Science, University of Nevada, Reno               |
| 2018–2020 | Visiting Professor of Computer Science, University of California, Davis           |
| 2015      | Research Intern, IBM Almaden Research Center, San Jose, CA                        |
| 2014      | Research Intern, Argonne National Laboratory, Chicago, IL                         |
| 2013      | Research Intern, Pacific Northwest National Laboratory, Seattle, WA               |
| 2009–2011 | Software Developer, Epic Systems Corporation, Madison, WI                         |
| 2006–2007 | Research Assistant, Centers for Disease Control and Prevention (CDC), Atlanta, GA |
| 2003–2004 | Research Assistant, Interuniversity Microelectronics Centre (IMEC), Belgium       |

## Awards and Grants

|      |                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | Federal Research Grant, NSF 2112345, \$255,916, PI at University of Nevada<br>“A DLT Machine Learning Platform for Blockchain Warehousing”                               |
| 2020 | Federal Research Grant, DOE DE-SC0020455, \$200,000, PI at University of Nevada<br>“HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”                      |
| 2019 | Best Paper Award, International Conference on Cloud Computing<br>“BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”                 |
| 2019 | Industrial Research Grant, Amazon Web Services, \$30,000                                                                                                                 |
| 2019 | Industrial Research Grant, Google Cloud Platform, \$15,000                                                                                                               |
| 2018 | Best Student Paper Award, IEEE International Conference on Cloud Computing<br>“Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory” |
| 2017 | Industrial Research Grant, Microsoft Azure, \$20,000                                                                                                                     |
| 2017 | Industrial Research Grant, Amazon Web Services, \$10,000                                                                                                                 |
| 2016 | Postdoctoral Fellowship, Sloan Foundation, \$155,000                                                                                                                     |
| 2007 | Graduate Fellowship, Oak Ridge Institute for Science and Education, \$85,000                                                                                             |

## Teaching

|                                     |                            |          |
|-------------------------------------|----------------------------|----------|
| CS 705: Cryptography and Blockchain | University of Nevada, Reno | 2022–now |
| CS 791: Distributed Computing       | University of Nevada, Reno | 2020–now |
| CS 457: Database Management Systems | University of Nevada, Reno | 2018–now |

## Services

|          |                        |                                                     |
|----------|------------------------|-----------------------------------------------------|
| 2021–Now | Editorial Review Board | IEEE Transactions on Distributed & Parallel Systems |
| 2023     | Program Committee      | AAAI-23                                             |
| 2022     | Grant Review Panel     | National Science Foundation: CISE OAC               |
| 2022     | Grant Review Panel     | National Science Foundation: CISE IIS               |
| 2019     | Organizing Committee   | ACM/IFIP International Middleware Conference        |
| 2018     | Grant Review Panel     | U.S. Department of Energy: ASCR                     |
| 2018     | Grant Review Panel     | National Science Foundation: CISE IIS               |

## Selected Publications

### Book chapters

- [1] **Dongfang Zhao**: Cross-Blockchain Transactions: Systems, Protocols, and Topological Theory. Handbook on Blockchain, Springer Optimization and its Applications. 2022.
- [2] **Dongfang Zhao**, Akash Mahakode, Sandip Lakshminarasaiah, Ioan Raicu: High-Performance Storage Support for Scientific Big Data Applications on the Cloud. Resource Management for Big Data Platforms 2016: 147-170

### Journals

- [1] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig: Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing. 8(2): 635-646 (2020)
- [2] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu: Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data 3(1): 107-117 (2017)
- [3] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu: Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems. 27(6): 1824-1837 (2016)
- [4] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu: Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Service Computing 9(1): 96-109 (2016)
- [5] **Dongfang Zhao**, Li Yang: Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence. 31(1): 86-98 (2009)

## Conferences

(<sup>†</sup>Student advised by **Dongfang Zhao**)

- [1] <sup>†</sup>Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [2] <sup>†</sup>Olamide Timothy Tawose, <sup>†</sup>Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022. Direct acceptance without revision
- [3] <sup>†</sup>Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [4] <sup>†</sup>Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [5] <sup>†</sup>Xinying Wang, <sup>†</sup>Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**: HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [6] <sup>†</sup>Abdullah Al-Mamun, Jun Dai, Xiaohua Xu, Mohammad Sadoghi, Haoting Shen, and **Dongfang Zhao**. DEAN: A Blockchain-Inspired Consensus Protocol Enabling Trustworthy Edge Computing. Network and Distributed System Security Symposium (NDSS). 2020.
- [7] Haoting Shen, Shahriar Badsha, and **Dongfang Zhao**. Poster: Consortium Blockchain for the Assurance of Supply Chain Security. Network and Distributed System Security Symposium (NDSS). 2020.
- [8] <sup>†</sup>Xinying Wang, <sup>†</sup>Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. The 12th International Conference on Cloud Computing (CLOUD), 2019. Best paper award
- [9] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan: Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [10] <sup>†</sup>Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**: Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD), 2018. Best student paper award
- [11] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad: Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment (VLDB). 10(11): 1226-1237 (2017)
- [12] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu: ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. The 27th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2013.